FEAT-ZD-48600 · BE-04 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Backend agent · item task.components · source hobasa-backend-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600

BE-04 "Identify the components/layers to be changed". This file lists the components and layers in ap_aging that the Backend tasks create or update, keeping each change as small as possible. It is based on a read of the current code and on the approved design. It adds no scope beyond TSK-05 (STORY-05) and TSK-06 (STORY-06).

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	BE-04 Identify the components/layers to be changed	record:brief
Owner / gate	Developer; automated checks, the independent judge and sampled human evaluation	record:brief
Inputs	feat.tasks (b0fd4d2), feat.stories (0590ea4), feat.prd (908b58d), feat.trd (3637e51), feat.design (722a98f), task.repo_access (daae080)	record:brief
Backend tasks in scope	TSK-05 (STORY-05, HC-6) and TSK-06 (STORY-06, HC-7), both in ap_aging	feat.tasks (section 1); task.repo_access (header)
Repository / branch	ap_aging, `feature/story` at `5d2cbb9`	task.repo_access (sections 1, 3)
Code read: source and tests	`src/ap_aging/aging.py`, `src/ap_aging/__init__.py`, `tests/test_aging.py`, `tests/__init__.py`	tool:repo
Code read: CI and docs	`ci.py`, `README.md`	file:ap_aging/ci.py; file:ap_aging/README.md

1. What the code does today

Component	Location	Behaviour today	Referenced by	Source
`BUCKETS`	`src/ap_aging/aging.py`	Tuple of the five bucket labels	`aging_buckets`	tool:repo
`days_past_due(due_date, as_of)`	`src/ap_aging/aging.py`	`max((as_of - due_date).days, 0)`: calendar days after the due date, or 0 when not yet due	`aging_buckets`; `tests/test_aging.py`	tool:repo
`bucket_for(days)`	`src/ap_aging/aging.py`	`days < 30` returns `"1-30"`, so exactly 30 days returns `"31-60"`	`aging_buckets`; `tests/test_aging.py`	tool:repo; feat.trd (F-10, R-1)
`aging_buckets(invoices, as_of)`	`src/ap_aging/aging.py`	Skips invoices where `inv.get("paid")` is truthy; sums `Decimal(inv["amount"])` per bucket	`src/ap_aging/__init__.py`; `tests/test_aging.py`	tool:repo
Typing import	`src/ap_aging/aging.py`	`from typing import Dict, Iterable` (no `List`)	—	tool:repo

Component	Location	Behaviour today	Referenced by	Source
Package interface	`src/ap_aging/__init__.py`	Imports `aging_buckets`; `__version__ = "1.4.0"`; `__all__ = ["aging_buckets"]`	Not named in any input	tool:repo
Existing tests	`tests/test_aging.py`	`test_days_past_due`, `test_bucket_edges`, `test_aging_buckets_skips_paid`, with `AS_OF = date(2026, 10, 25)`	—	tool:repo
CI	`ci.py`	Compiles the code and runs `unittest` discovery under `tests/`. Requires coverage of `src/ap_aging` of at least `COVERAGE_MIN = 90.0`. Static checks flag `print`/`eval`/`exec`, bare `except`, and lines over 140 characters	—	file:ap_aging/ci.py

The code has no service or transport layer between ap_aging and ap_portal, and no database. `src/ap_aging/` holds only `aging.py` and `__init__.py` (tool:repo; feat.design A3 "Transport for CT-1"; task.repo_access section 4).

Items marked **agent judgement** below are this agent's proposals for the smallest change. No input states them.

2. Layers: changed or not

Layer	Change	Why	Source
Domain logic (`src/ap_aging/aging.py`)	**Update (additive only)**	Both new functions go here, next to `days_past_due`, which `overdue_with_risk` reuses	feat.design (DD-2, B3.1); feat.trd (C-6, C-7)
Package public interface (`src/ap_aging/__init__.py`)	**Update**	Export the two new functions and make one minor version step	feat.design (B3.1, DD-11); feat.tasks (section 2)
Unit tests (`tests/`)	**Create** `tests/test_overdue.py`	New file for the TSK-05 and TSK-06 tests; `tests/test_aging.py` stays unchanged	feat.tasks (section 2, TSK-05, TSK-06)
Serialisation (CT-1)	Inside the domain module (`overdue_rows_to_json`); no new layer	ap_aging owns serialisation; no transport is added	feat.design (DD-3, CT-1)
Transport / API / service	**No change**	Adding a transport would reopen DES-01	feat.design (DD-3, A8); feat.tasks (TSK-06 "Must not change")
Database / schema	**No change**	No repository uses a database	task.repo_access (section 4); feat.design (A1 "Data")
Dependencies	**No change**	Standard library only (`json`, `decimal`, `datetime`, `typing`)	feat.design (DD-1); file:ap_aging/CLAUDE.md
CI (`ci.py`)	**No change**	Discovery already picks up any new test file under `tests/`, and the 90% gate covers the new code	file:ap_aging/ci.py

Layer	Change	Why	Source
Docs (`README.md`)	**No change**	The README does not list functions, so nothing in it goes out of date	file:ap_aging/README.md

3. Components to create or update (minimum change)

3.1 `src/ap_aging/aging.py` *(update, additive)*

#	Change	Task	Minimal form	Source
A1	Add `import json`	TSK-06	One import line at the top of the module	feat.design (B3.1); feat.tasks (TSK-06)
A2	Add `List` to the typing import	TSK-05	Change the line to `from typing import Dict, Iterable, List`. Agent judgement: extend the existing line rather than add a new one	tool:repo; feat.design (B3.1)
A3	New constant `OVERDUE_AFTER_DAYS = 30`	TSK-05	Agent judgement: one line next to `BUCKETS`	feat.design (B3.2, DD-4)
A4	New constant `TWO_PLACES = Decimal("0.01")`	TSK-05 (TSK-06 reuses it)	Agent judgement: one line next to `OVERDUE_AFTER_DAYS`	feat.design (B3.2); feat.tasks (TSK-06 notes)
A5	New function `overdue_with_risk(invoices: Iterable[Dict], as_of: date, risk_by_vendor: Dict[str, str]) -> List[Dict]`	TSK-05	One loop. Skip the invoice when `inv.get("paid")` is truthy. Set `days = days_past_due(inv["due_date"], as_of)` and keep only `days > OVERDUE_AFTER_DAYS`. Build the five-key row with `Decimal(inv["amount"]).quantize(TWO_PLACES)` and `risk_by_vendor.get(inv["vendor"], "unknown")`. Then `rows.sort(key=lambda row: (-row["days"], row["id"]))`. Docstring as in feat.design B3.2. Agent judgement: no private helpers are needed, although size class M would allow them	feat.design (B3.2 steps 1 to 4); feat.trd (IF-5.1 to IF-5.7); feat.tasks (section 2, class M)
A6	New function `overdue_rows_to_json(rows: Iterable[Dict]) -> str`	TSK-06	A single `json.dumps` over a list comprehension, in CT-1 key order: `id`, `vendor`, `days` as `int`, `amount` as `str(Decimal(r["amount"]).quantize(TWO_PLACES))`, `risk`. Docstring as in feat.design B3.2	feat.design (B3.2, CT-1); feat.tasks (TSK-06 notes)

Reused, not changed:

- `days_past_due`, which counts calendar days (D-2026-031).

- The paid test `inv.get("paid")`, which `aging_buckets` already uses (D-2026-050).

Sources: feat.design (NF-2, NF-3); tool:repo; record:memory.

Must not change: `BUCKETS`, `days_past_due`, `bucket_for` and `aging_buckets`, including their signatures and results. `overdue_with_risk` must **never call** `bucket_for`: `bucket_for(30)` returns `"31-60"`, while AC-05.2 excludes an invoice exactly 30 days past due. Sources: feat.tasks (TSK-05); feat.design (DD-4); feat.trd (F-10); tool:repo.

Rules the code must keep:

- Money stays `Decimal` and never passes through `float` (D-2026-044, L-011; feat.design NF-1).
- `as_of` is required, with no default (feat.tasks TSK-05 notes).
- A missing `vendor` on an overdue unpaid invoice raises `KeyError`, with no silent default (feat.tasks TSK-05 notes; feat.design B3.4).
- Both functions are pure: no I/O and no change to their inputs (feat.design NF-10).

Sources: feat.tasks (TSK-05 notes); feat.design (NF-1, NF-10, B3.4); record:memory.

3.2 `src/ap_aging/__init__.py` *(update)*

#	Change	Task	Minimal form	Source
I1	Import the new functions	TSK-05, TSK-06	Agent judgement: extend the existing line to `from .aging import aging_buckets, overdue_rows_to_json, overdue_with_risk`	tool:repo; feat.design (B3.1)
I2	Add them to `__all__`	TSK-05, TSK-06	`__all__ = ["aging_buckets", "overdue_with_risk", "overdue_rows_to_json"]`	feat.trd (C-6); feat.design (B3.1)
I3	Version step	TSK-05 (owns the ap_aging version step)	One minor step from `"1.4.0"`. The Developer confirms the exact number (open question BC-2)	feat.tasks (section 2); feat.design (DD-11); tool:repo

3.3 `tests/test_overdue.py` *(create)*

Setup:

- Use `AS_OF = date(2026, 10, 25)`.
- Build due dates as `AS_OF - timedelta(days=N)` so that `days` is 30, 31, 42 and 75.
- Use synthetic data only (`SYN-INV-*`, "Synthetic Vendor <letter>").

Sources: feat.tasks (TSK-05); feat.design (B3.5, SE-4).

Test	Covers	Task	Source
`test_overdue_boundary_30_and_31`	AC-05.1, AC-05.2 (L-007)	TSK-05	feat.tasks (TSK-05); record:memory
`test_overdue_excludes_paid_and_not_due`	AC-05.3, AC-05.4	TSK-05	feat.tasks (TSK-05)
`test_overdue_row_keys_and_sort`	AC-05.5, AC-05.6 (tie on 42: `SYN-INV-002` before `SYN-INV-003`)	TSK-05	feat.tasks (TSK-05); feat.stories (STORY-05)

Test	Covers	Task	Source
`test_overdue_unknown_risk`	AC-05.7	TSK-05	feat.tasks (TSK-05)
`test_overdue_amount_quantised`	AC-05.8 (`Decimal("980.5")` → `Decimal("980.50")`)	TSK-05	feat.tasks (TSK-05)
`test_overdue_missing_vendor_raises`	AC-05.10	TSK-05	feat.tasks (TSK-05)
`test_existing_aging_unchanged`	AC-05.9 (pins `bucket_for(30) == "31-60"`)	TSK-05	feat.tasks (TSK-05); feat.design (B3.5)
`test_rows_to_json`	AC-06.1 to AC-06.3: `"1250.00"` for SYN-INV-004; every amount a two-decimal string; `days` an integer; `[]` for no rows; CT-1 key order. Compare `json.loads` values, not raw strings. Agent judgement: also assert that no row has a date field, to cover AC-06.4 from feat.stories, which the TSK-06 test list does not name	TSK-06	feat.tasks (TSK-06); feat.design (B3.5); feat.stories (AC-06.4)

`tests/test_aging.py` stays unchanged (feat.tasks section 2), and so does `tests/__init__.py` (tool:repo).

4. Order of work

Order	Change	Depends on	Source
1	TSK-05: A2, A3, A4, A5; I1 and I2 for `overdue_with_risk`; I3; the TSK-05 tests	Nothing in code	feat.tasks (TSK-05)
2	TSK-06: A1, A6; I1 and I2 for `overdue_rows_to_json`; `test_rows_to_json`	TSK-05: takes its rows and reuses `TWO_PLACES`	feat.tasks (TSK-06); feat.stories (section 2)

Neither task depends on another lane's code. ap_aging reads plain dicts (DC-1, DC-3) and imports no other package (feat.design Part C "Lane dependencies").

5. Why this is the smallest change

- **Only additive code.** Four existing lines change: the typing import in `aging.py`, and the import, `__version__` and `__all__` lines in `__init__.py`. No existing signature changes (file:ap_aging/CLAUDE.md; feat.design DD-2, NF-4; tool:repo).
- **No new module, layer or dependency.** Agent judgement: no private helper either. Both functions sit in `aging.py`, next to the `days_past_due` they reuse (feat.design DD-1, DD-2; tool:repo).
- **Two new constants only.** `overdue_with_risk` uses both `OVERDUE_AFTER_DAYS` and `TWO_PLACES`; `overdue_rows_to_json` reuses `TWO_PLACES` (feat.design B3.2).
- **No explicit error handling.** A missing `vendor` raises `KeyError` through `inv["vendor"]`, and `test_overdue_missing_vendor_raises` covers it (feat.design B3.4). Agent judgement: other missing keys also raise `KeyError` through indexing, and no input asks for them to be handled.
- **No CI or README change** (file:ap_aging/ci.py; file:ap_aging/README.md).

6. Risks for later steps

ID	Risk	Handling	Source
CR-1	Reusing `bucket_for` would wrongly list an invoice that is exactly 30 days past due	Use `OVERDUE_AFTER_DAYS` with strict >; covered by `test_overdue_boundary_30_and_31`	feat.trd (F-10, R-1); feat.design (DD-4)
CR-2	Coverage could fall below the 90% gate	Agent judgement: the tests in section 3.3 reach the paid, not-yet-due, exactly-30, over-30, unknown-risk, missing-vendor and empty-list paths. BE-06 measures the actual figure	file:ap_aging/ci.py; feat.tasks (section 2 "CI bar")
CR-3	CI imports the third-party `coverage` package, and nobody has checked that it is installed	Carried from task.repo_access RA-2	task.repo_access (section 5, RA-2)

7. Open questions (non-blocking)

ID	Question	Interim handling	Owner (proposed)
BC-1	Where should the new tests go? feat.stories TK-05.2 suggests `tests/test_aging.py`. The later feat.tasks (section 2) assigns a new `tests/test_overdue.py`. feat.design B3.5 allows either	Follow feat.tasks: create `tests/test_overdue.py`	Developer
BC-2	What is the exact ap_aging version after the minor step from `1.4.0`? No input gives the number	Propose a minor step in BE-05 for the Developer to confirm	Developer
—	Carried unchanged: RA-1 (one shared branch, `feature/story`, for STORY-05 and STORY-06); RA-2 (whether `coverage` is installed); TQ-1 (`bucket_for(30)`); TQ-3 (CT-1 approval; AC-06.1 to AC-06.4 pending); TQ-9 (`KeyError` for a missing vendor; AC-05.10 pending); PQ-2 (sort order and the meaning of `days`; AC-05.6 pending); DQ-2 (rounding mode)	As recorded in the source files	As recorded

Sources for the carried items: task.repo_access (section 6); feat.trd (section 7.2); feat.stories (STORY-05, STORY-06); feat.design (D3).

8. Validation check

Check	Result	Source
Existing code read and understood	Pass (section 1)	tool:repo; file:ap_aging/ci.py; file:ap_aging/README.md
Every component to create or update is listed with its task	Pass (section 3)	feat.tasks (TSK-05, TSK-06)
The change matches the design	Pass (HC-6, HC-7, B3.1 to B3.5, DD-3, DD-4)	feat.design
No existing signature changes; additive only	Pass	file:ap_aging/CLAUDE.md; feat.design (NF-4)

Check	Result	Source
Decisions D-2026-031, D-2026-044, D-2026-050; lessons L-007, L-011	Consistent	record:memory
Database or dependency change needed	None	task.repo_access (section 4); feat.design (DD-1)
Blocking questions	None	—